

Razma and Allan Group 310

Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In this lab, we learned how to build different logic circuits using only NAND gates. We practiced DeMorgan's Law and how to use it to convert AND and OR gates into NAND gates. We also learned that a NAND gate can work like a NOT gate when both of its inputs are connected. We used these ideas to build the given logic circuit on a breadboard and test it.

Lab Questions

1 - Write down DeMorgan's Law and the truth tables proving it out.

Lab = 3

4 NAND chip.

$$F = (A \& \neg B) \mid (C \& D)$$

$$F = (A \cdot \bar{B}) + (C \cdot D)$$

$\underbrace{\quad}_x \qquad \underbrace{\quad}_y$

De Morgan's Law

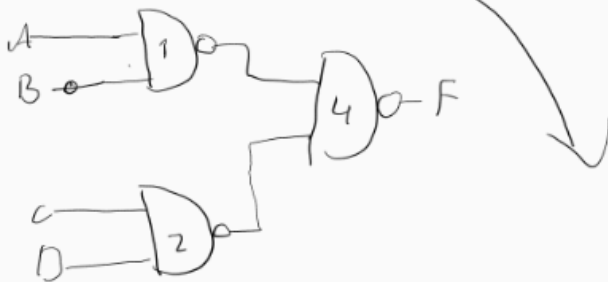
De Morgan's Law:

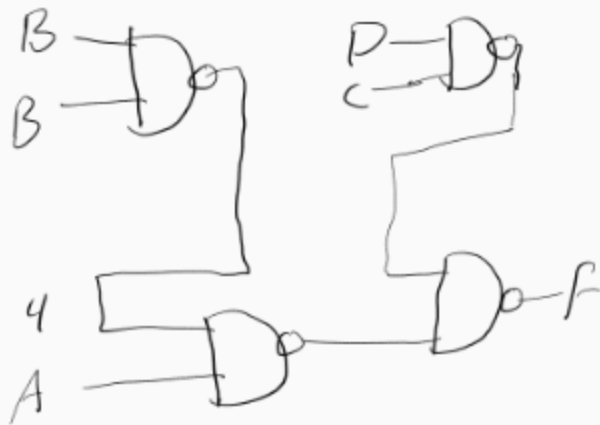
$$\overline{(A \cdot B)} = (\bar{A} + \bar{B})$$

$$x + y = \overline{\bar{x} \cdot \bar{y}}$$

$$\overline{(A + B)} = (\bar{A} \cdot \bar{B})$$

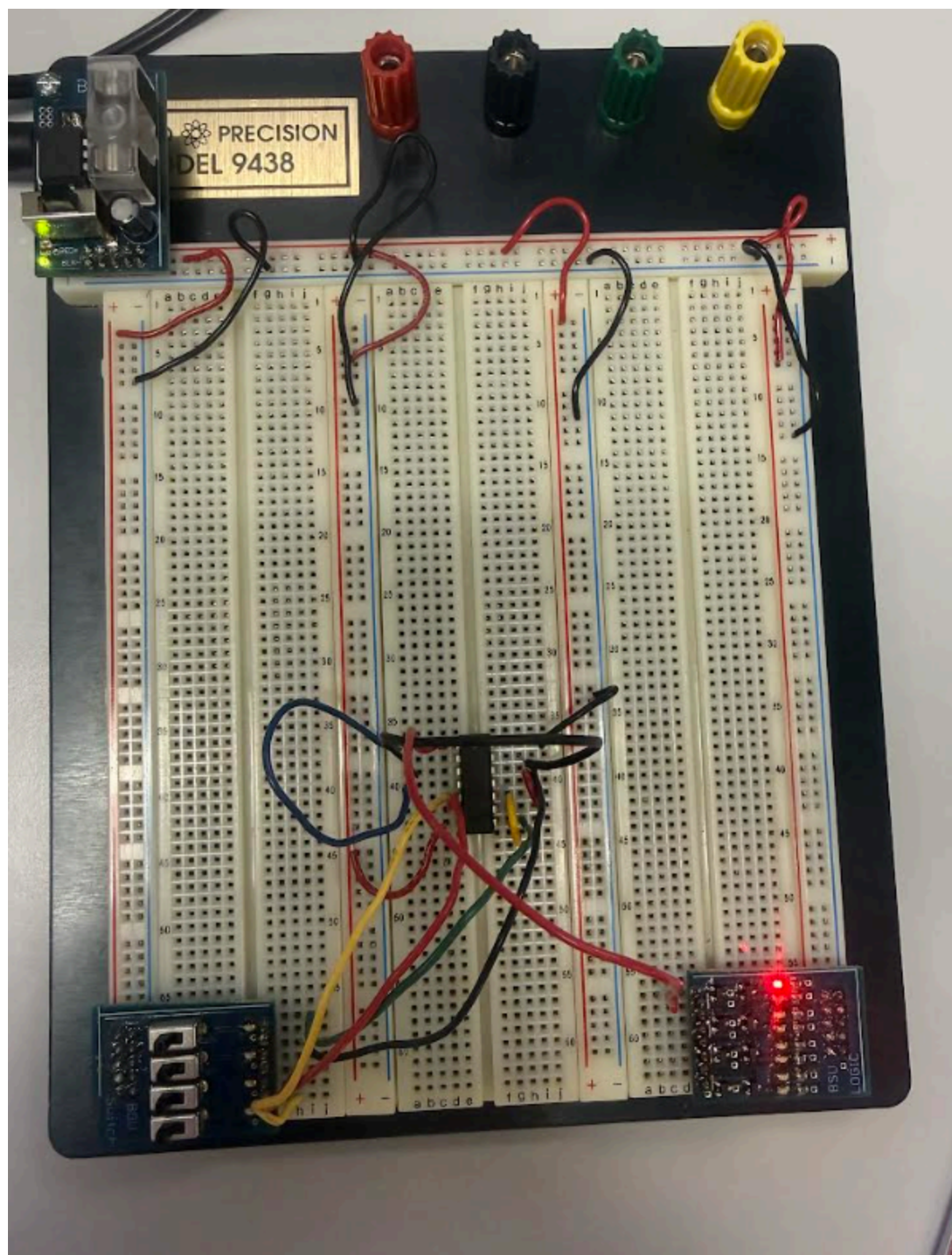
$$F = \overline{\overline{(A \cdot \bar{B})} \cdot \overline{(C \cdot D)}} \quad \text{4 NAND}$$





C and D

A	B	C	D	$A \& \sim B$	$C \& D$	$(A \& \sim B) (C \& D)$
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	1	1
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	1	0	1
1	0	0	1	1	0	1
1	0	1	0	1	0	1
1	0	1	1	1	1	1
1	1	0	0	0	0	0
1	1	0	1	0	0	0
1	1	1	0	0	0	0
1	1	1	1	0	1	1



2 - What is the value in converting circuits to NAND only?

One benefit is that we can build different types of circuits using only one type of gate. This can make the design simpler because we do not need separate AND, OR, and NOT gates. NAND gates can be connected in different ways to do the same jobs as other gates.

3 - How does what you did in lab with the breadboard relate to the FPGA?

The breadboard helped us understand how a logic circuit works by actually building it and testing it. We connected the NAND gates and wires to make the equation and see the output. This relates to an FPGA because an FPGA also uses logic gates to create circuits, but it does everything inside the chip. This lab helped me understand how the logic we build on a breadboard can also be used in an FPGA.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.